

## Ideal Projectile Motion

Date\_\_\_\_\_ Period\_\_\_\_

- 1) A soccer ball is kicked across a field at an angle of  $30^\circ$  with an initial speed of 16 m/s. Write a set of parametric equations for the motion of the soccer ball.

$$x = 8t\sqrt{3}, \quad y = 8t - \frac{9.8t^2}{2}$$

- 2) A golf ball is struck across a flat fairway at an angle of  $45^\circ$  with an initial speed of 136 ft/s. Write a set of parametric equations for the motion of the golf ball.

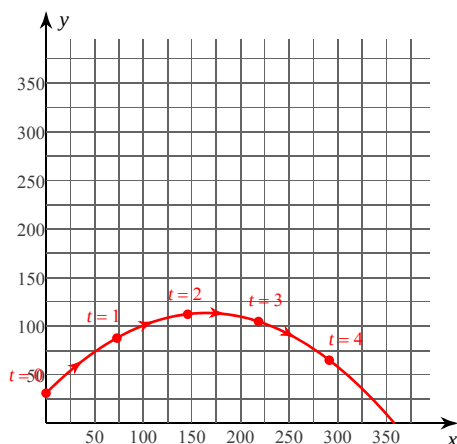
$$x = 68t\sqrt{2}, \quad y = 68t\sqrt{2} - \frac{32.17t^2}{2}$$

- 3) A snowball is thrown at an angle of  $60^\circ$  with an initial speed of 45 ft/s and an initial height of 5 ft. Write a set of parametric equations for the motion of the snowball.

$$x = \frac{45t}{2}, \quad y = 5 - \frac{32.17t^2}{2} + \frac{45t\sqrt{3}}{2}$$

- 4) A cannon ball is fired across a flat field at an angle of  $45^\circ$  with an initial speed of 103 ft/s and an initial height of 31 ft.

- Write a set of parametric equations for the motion of the cannon ball.
- Determine how long the cannon ball was in the air.
- Determine how far the cannon ball traveled in the air.
- Determine when the cannon ball reached its maximum height.
- Determine the maximum height reached by the cannon ball.
- Sketch the graph of the parametric equations.



$$\text{i) } x = \frac{103t\sqrt{2}}{2}, \quad y = 31 - \frac{32.17t^2}{2} + \frac{103t\sqrt{2}}{2}$$

$$\text{ii) } 4.92 \text{ s}$$

$$\text{iii) } 358.31 \text{ ft}$$

$$\text{iv) } 2.26 \text{ s}$$

$$\text{v) } 113.44 \text{ ft}$$